4. Crypt-Arithmetic Problem

Aim

To solve the crypt-arithmetic puzzle SEND + MORE = MONEY, where each letter represents a unique digit.

Algorithm

1. Import permutations to generate possible digit assignments.
2. Assign different digits from 0 to 9 to the letters S, E, N, D, M, O, R and Y.
3. Reject assignments in which S or M is zero because they are leading digits.
4. Convert SEND, MORE and MONEY into numbers.
5. Check whether SEND + MORE equals MONEY.
6. Print the first valid solution and stop.

Program

untitled (3.13.15)

File Edit Format Run Options Window Help

```
from itertools import permutations

letters = "SENDMORY"
for p in permutations(range(10), len(letters)):
    d = dict(zip(letters, p))

    if d["S"] == 0 or d["M"] == 0:
        continue

    send = 1000*d["S"] + 100*d["E"] + 10*d["N"] + d["D"]
    more = 1000*d["M"] + 100*d["O"] + 10*d["R"] + d["E"]
    money = (10000*d["M"] + 1000*d["O"] + 100*d["N"]
             + 10*d["E"] + d["Y"])

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Output

IDLE Shell 3.13.15

File Edit Shell Debug Options Window Help

```
Python 3.13.15 (tags/v3.13.15:4061bc4,
 (AMD64)] on win32
Enter "help" below or click "Help" abov
>>>
==== RESTART: C:/Users/Barath/AppData/L
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```

Result

Thus, a valid solution to SEND + MORE = MONEY was found: 9567 + 1085 = 10652.